DEPARTMENT OF PHYSICS

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Master's Thesis

Simulation of Radiative Effects for the AMBER Proton Radius Measurement with McMule

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Abstract

Dieses Projekt ist ganz toll, ich schwör.

It is remarkable that all ordinary visible matter is composed of only three stable fundamental particles: the electron, the up quark, and the down quark. While other elementary particles, such as photons, gluons, and the weak gauge bosons, play essential roles in mediating the fundamental interactions, they do not form stable matter themselves. Despite the remarkable success of the Standard Model, it is widely recognized as an incomplete theory, as it cannot explain several fundamental phenomena, such as the nature of dark matter, the origin and ordering of neutrino masses, or the matter-antimatter asymmetry observed in the Universe. Among the many open questions in particle physics is the precise understanding of the proton's internal structure. The so-called proton radius puzzle emerged from conflicting measurements of the proton's charge radius obtained using different experimental approaches. Although recent experimental results have substantially reduced this discrepancy, an independent high-precision determination of the proton radius remains of great importance. The AMBER experiment at CERN aims to contribute to this effort by measuring the proton charge radius with unprecedented precision.

Preface

Ob das so nötig ist, ist die Frage…

Contents

1 Introduction: The Proton Radius Puzzle

The proton is not a point-like particle, but consists of quarks and gluons bound together by the strong interaction. As a result, its electric charge is distributed over a finite volume rather than concentrated at a single point. This finite spatial extent is quantified by the proton charge radius.

The internal structure of the proton becomes relevant when it is probed by a lepton. Instead of interacting with a point-like charge, the lepton probes the proton's extended charge distribution, resulting in measurable effects. These effects are described by electromagnetic form factors, which parameterize the proton's internal charge and magnetization distributions and therefore contain information about its spatial structure. In particular, the electric Sachs form factor $G_E(Q^2)$ characterizes the distribution of the proton's electric charge. Its slope at zero momentum transfer is directly related to the proton charge radius,

$$r_p^2 = -6 \left| \frac{dG_E(Q^2)}{dQ^2} \right|_{Q^2=0} .$$

The proton charge radius can be determined using different experimental approaches. In scattering experiments, leptons are scattered off protons and the proton structure is extracted from the dependence of the scattering cross section on the momentum transfer Q^2. Since the momentum transfer corresponds to the spatial resolution of the probe, measurements at different Q^2 values provide information about the proton's charge distribution through the electromagnetic form factors. The relation between the scattering process, the momentum transfer, and the form factors will be discussed in more detail in Chapter **??**.

Another approach is based on spectroscopy, which determines atomic energy differences by measuring the absorption or emission of electromagnetic radiation. A particularly sensitive observable is the Lamb shift. In the Dirac description of hydrogen with a point-like proton, the $2S_{1/2}$ and $2P_{1/2}$ states are degenerate. However, in 1947, Lamb and Retherford experimentally demonstrated that the $2S_{1/2}$ state lies slightly above the $2P_{1/2}$ state in energy. This observation played an important role in the development of quantum electrodynamics (QED), which explains the Lamb shift through radiative corrections arising from the interaction of the lepton with the quantized electromagnetic field. These corrections are described by loop diagrams involving virtual particles. Since the two states have different wave functions near the proton, they are affected differently, resulting in a measurable energy difference.

Measurements of the Lamb shift in muonic hydrogen provide an especially precise determination of the proton radius. Muonic hydrogen is a hydrogen-like atom in which the electron is replaced by a negatively charged muon. Due to the much larger mass of the muon compared to the electron, the Bohr radius is approximately 200 times smaller than in ordinary hydrogen. As a result, the muon has a significantly higher probability density near the proton, making the energy levels much more sensitive to the finite size of the proton.

Historically, measurements based on elastic electron-proton scattering and spectroscopy of ordinary hydrogen yielded a proton charge radius of approximately 0.88 fm. The high precision achieved with muonic hydrogen spectroscopy led to a significantly smaller value of approximately 0.84 fm, which was inconsistent with previous measurements. This discrepancy, known as the proton radius puzzle, triggered extensive theoretical and experimental investigations. Although recent measurements have substantially reduced the discrepancy, an independent high-precision determination of the proton radius remains essential.

2 The AMBER Experiment

2.1 Overview of the AMBER Experiment

The AMBER (Apparatus for Meson and Baryon Experimental Research) experiment, designated NA66, is a fixed-target experiment at CERN. It is located at the Super Proton Synchrotron (SPS) in the North Area of CERN, shown in 2.1. It was approved in 2020 as a multi-purpose facility dedicated to precision studies in hadron physics. Building upon the spectrometer and infrastructure of the former COMPASS experiment, AMBER combines existing detector systems with substantial upgrades to address a broad physics program. It can be operated with both muon and hadron beams. These come from the M2 beam line, which is directly connected to the SPS.

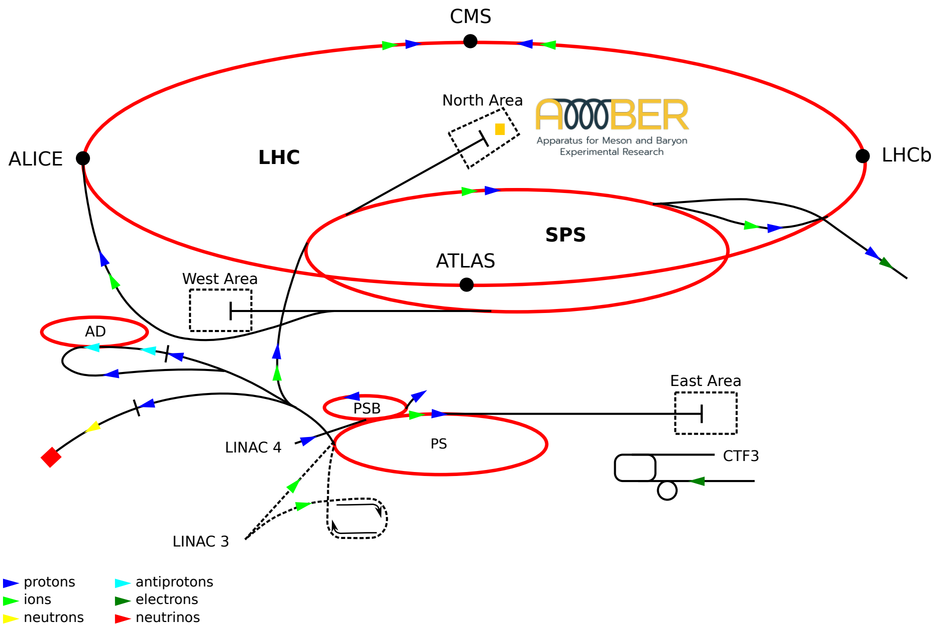

Figure 2.1: Location of the AMBER experiment within the CERN accelerator complex. Figure adapted from [1].

In Phase-1 (2022-2026), the experiment focuses on measurements of antiproton production cross sections, investigations of the partonic structure of mesons through Drell-Yan processes, hadron spectroscopy, and a high-precision proton radius measurement (PRM). AMBER aims to determine the proton charge radius using elastic muon-proton scattering. The measurement covers a very low four-momentum-transfer region, where the electric form factor is particularly sensitive to the proton's charge radius.

After Long Shutdown 3, in Phase-2, AMBER plans to extend its physics program towards measurements requiring high-intensity hadron beams. The main objectives include precision studies of the internal structure of hadrons, such as the three-dimensional partonic structure of mesons and baryons, as well as further investigations of hadron production mechanisms.

2.2 Experimental Setup for the Proton Radius Measurement

The proton-radius measurement is performed using a 100 GeV muon beam. The beam is delivered to the detector in spills. A spill refers to a single period during which the beam is extracted from the accelerator and delivered to the experiment. For AMBER, a spill is approximately 4.8 s long. Therefore, an intensity of approximately 4×10^6 muons per spill corresponds to an average rate of about 8.3×10^5 muons per second that hit the target during the spill. The beam is provided by the M2 beam line and is directed onto an active hydrogen target operated at a pressure of 20 bar. The high pressure increases the density of hydrogen atoms in the target, thereby increasing the probability of elastic muon-proton scattering. Since hydrogen provides a clean proton target, it allows the proton kinematics to be reconstructed without additional nuclear effects. The scattered muon is measured with the COMPASS spectrometer, while the recoil proton is reconstructed inside the target using a Time Projection Chamber (TPC). An overview of the experimental setup is shown in Figure 2.2.

The direction of the incoming beam defines the longitudinal axis of the experiment. Detector components located before the interaction point with respect to the beam direction are referred to as upstream, whereas components located after the interaction point are called downstream.

The main detector components relevant for the PRM are described in the following.

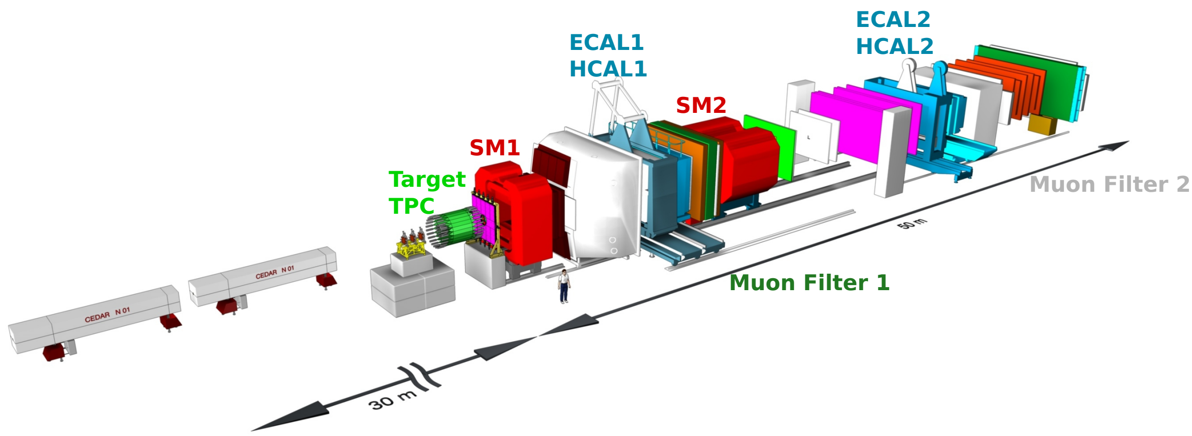

Figure 2.2: LALALA. Figure adapted from [2].

Combined Tracking Stations Four combined tracking stations surround the target region, with two stations located upstream and two downstream of the TPC. Their primary purpose is to reconstruct the trajectories of the incoming and scattered muons with high spatial and temporal precision.

Each tracking station combines a silicon pixel detector with several layers of scintillating fibers. The pixel detector provides excellent spatial resolution for determining the interaction vertex and scattering angles. The scintillating fibers, consisting of thin plastic fiber ribbons connected to photomultipliers, offer a time resolution of less than 1 ns. This precise timing is crucial to disentangle individual beam particles and match track segments with the TPC signals, where the total electron drift time is on the order of 100 µs.

Time Projection Chamber The TPC serves as an active hydrogen target. It is 2.3 m long and operated with hydrogen gas at a pressure of 20 bar. The chamber consists of four cylindrical drift cells, each with a length of 40 cm and a diameter of 60 cm.

When a charged particle traverses the hydrogen gas, it ionizes the gas molecules and creates electron-ion pairs along its trajectory. An electric drift field causes the liberated electrons to move towards the readout plane, where their arrival time and position are measured. From these signals, the three-dimensional trajectory of the recoil proton can be reconstructed.

Since the proton loses energy as it traverses the gas, its kinetic energy can be determined from the measured ionization. Together with the reconstructed momentum, this allows the proton kinematics and the squared four-momentum transfer Q^2 to be determined with high precision.

Spectrometer Magnets Downstream of the target, the spectrometer consists of two dipole magnets, SM1 and SM2. During the PRM, the first magnet (SM1) is switched off to prevent small-angle scattered muons from being deflected before reaching the downstream target tracking stations. Consequently, momentum reconstruction for the scattered muon is performed exclusively using the second dipole magnet, SM2, and the measurements of the tracking detectors.

This single-magnet deflection scheme is well-suited for elastic muon-proton scattering at low Q^2, as the process is strongly forward-peaked, giving rise to small scattering angles and high forward momenta.

Electromagnetic and Hadronic Calorimeters The spectrometer features a two-stage calorimeter system containing two electromagnetic calorimeters (ECAL1 and ECAL2) and two hadronic calorimeters (HCAL1 and HCAL2). Calorimeters measure the energy of particles through the detection of the energy deposited in active detector material.

Electromagnetic calorimeters are designed to absorb and measure the energy of light particles, primarily electrons, positrons, and photons. High-energy electrons lose energy via Bremsstrahlung, while photons convert into electron-positron pairs. This alternating process initiates an electromagnetic shower characterized by the material's radiation length. The total deposited energy is related to the incident particle energy and can be used to reconstruct photons and electrons.

The AMBER setup employs lead-glass blocks and sampling Shashlik-type modules (alternating layers of lead absorber and plastic scintillator). The light generated by Čerenkov radiation or scintillation inside the absorber material is collected by photodetectors (such as photomultipliers or avalanche photodiodes) and converted into an electrical signal proportional to the particle's energy.

In the context of the PRM, the electromagnetic calorimeters are particularly important because photons emitted during radiative muon-proton scattering have to be detected and studied. These photons modify the experimentally reconstructed kinematics and therefore play an important role in the determination of radiative corrections. In particular, ECAL2 provides information on photons emitted at larger distances downstream of the target and is an essential detector for the studies presented in this thesis.

Hadronic calorimeters measure the energy of strongly interacting particles (hadron species such as protons, pions, and kaons). Hadrons interact with atomic nuclei inside dense absorber materials (iron or lead), generating a hadronic cascade. Although they are not the primary detectors for the proton-radius measurement, they contribute to particle identification and background suppression within the spectrometer.

Muon Filters To ensure unambiguous identification of scattered muons, the spectrometer incorporates two dedicated muon filter systems: MF1 (upstream) and MF2 (downstream).

Each muon filter consists of a thick iron or concrete absorber wall followed by tracking detectors. Strongly interacting hadrons and remaining electromagnetic shower components undergo nuclear interactions or showers and are completely absorbed within the material. In contrast, high-energy muons behave as minimum ionizing particles, penetrating the absorber material with minimal energy loss.

By matching tracks recorded in the tracking detectors directly behind the absorber walls with momentum tracks reconstructed via SM2, MF1 and MF2 provide a clean and highly efficient identification of the scattered muon.

3 Theoretical Background on Elastic Muon-Proton Scattering

3.1 From transition rates to scattering cross sections

The transition rate of a quantum mechanical process is described by Fermi's golden rule. It relates the probability of a transition between an initial state and a final state to the interaction matrix element and the available phase space:

$$\Gamma_{i\to f} = \frac{2\pi}{\hbar} \left|\mathcal{M}_{fi}\right|^2 \rho(E_f), \tag{3.1}$$

where $\mathcal{M}_{fi}$ is the matrix element of the interaction and contains all information about the dynamics of the process. It is calculated from the corresponding Feynman diagrams. The factor $\rho(E_f)$ describes the density of accessible final states and contains only kinematic information. It depends on the masses, energies, and momenta of the particles involved. A larger available phase space increases the number of possible final states and therefore enhances the transition probability.

For relativistic processes, the phase space is expressed in a Lorentz-invariant form. For a process with n particles in the final state, the differential transition rate is given by

$$d\Gamma = \frac{(2\pi)^4}{2E_1}|\mathcal{M}|^2\delta^{(4)}\left(p_1 - \sum_f p_f\right) d\Phi_n, \tag{3.2}$$

where the Lorentz-invariant phase space element is defined as

$$d\Phi_n = \prod_f \frac{d^3\mathbf{p}_f}{(2\pi)^3 2E_f}. \tag{3.3}$$

The four-dimensional delta function ensures conservation of energy and momentum. In scattering processes, the relevant observable is not the transition rate but the cross section. It is obtained by normalizing the transition rate to the incident particle flux,

$$d\sigma = \frac{d\Gamma}{\text{flux}}. \tag{3.4}$$

The flux factor accounts for the relative motion of the incoming particles and normalizes the transition rate to the number of possible collisions. For a general $2 \to n$ scattering process, this results in the Lorentz-invariant expression

$$d\sigma = \frac{(2\pi)^4|\mathcal{M}|^2}{4\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}} \delta^{(4)} \left(p_1 + p_2 - \sum_f p_f \right) d\Phi_n. \tag{3.5}$$

3.2 Non-Radiative Kinematics

In this thesis, only elastic muon-proton scattering is considered,

$$p(p_1) + \mu(p_2) \to p(p_3) + \mu(p_4). \tag{3.6}$$

The proton is initially at rest, therefore the incoming four-momenta are given by

$$p_1 = (m_p, \mathbf{0}), \qquad p_2 = (E_\mu, \mathbf{p}_\mu), \tag{3.7}$$

where m_p is the proton mass and E_μ is the energy of the incoming muon. The outgoing proton and muon are described by

$$p_3 = (E'_p, \mathbf{p}'_p), \qquad p_4 = (E'_\mu, \mathbf{p}'_\mu). \tag{3.8}$$

Energy and momentum conservation require

$$p_1 + p_2 = p_3 + p_4. \tag{3.9}$$

The interaction is mediated by the exchange of a virtual photon with four-momentum

$$q = p_2 - p_4 = p_3 - p_1. \tag{3.10}$$

The invariant momentum transfer is defined as

$$Q^2 = -q^2, \tag{3.11}$$

where the positive sign convention for Q^2 is commonly used in scattering experiments. The value of Q^2 determines the resolution scale of the interaction: larger values of Q^2 correspond to a smaller distance scale probed inside the proton.

To describe relativistic scattering processes, it is convenient to use the Lorentz-invariant Mandelstam variables. Since they are independent of the reference frame, they provide a compact and universal way to express scattering kinematics and cross sections. For a $2 \to 2$ scattering process, the kinematics can be described by

$$\begin{aligned} s &= (p_1 + p_2)^2 = (p_3 + p_4)^2 = m_p^2 + 2m_p E_\mu + m_\mu^2, \\ t &= (p_1 - p_3)^2 = (p_2 - p_4)^2 = 2m_p^2 - 2m_p E'_p = -Q^2, \\ u &= (p_1 - p_4)^2 = (p_2 - p_3)^2 = m_p^2 - 2m_p E'_\mu + m_\mu^2. \end{aligned} \tag{3.12}$$

They satisfy

$$s + t + u = 2m_p^2 + 2m_\mu^2.$$

3.3 Differential Cross Section for Elastic Muon-Proton Scattering

The differential cross section for elastic muon-proton scattering can be expressed in terms of the Mandelstam variables and the matrix element of the interaction. The general expression for a two-particle scattering process is given by

$$\frac{d\sigma}{dt} = \frac{1}{64\pi s}\frac{1}{|\mathbf{p}_{1,\mathrm{cm}}|^2}|\mathcal{M}|^2,$$

where $|\mathbf{p}_{1,\mathrm{cm}}|$ is the magnitude of the incoming particle momentum in the center-of-mass frame. To express the momentum in terms of the Mandelstam variable s, the Källén function is introduced:

$$\lambda(s, m_\mu^2, m_p^2) = s^2 + m_\mu^4 + m_p^4 - 2s(m_\mu^2 + m_p^2) - 2m_\mu^2 m_p^2.$$

The center-of-mass momentum is related to the Källén function by

$$|\mathbf{p}_{1,\mathrm{cm}}|^2 = \frac{\lambda(s, m_\mu^2, m_p^2)}{4s}.$$

Therefore, the differential cross section can be rewritten as

$$\frac{d\sigma}{dt} = \frac{1}{16\pi\lambda(s, m_\mu^2, m_p^2)}|\mathcal{M}|^2.$$

This expression is written in a Lorentz-invariant form and can therefore be evaluated in any reference frame.

However, the experimentally measured quantity is not directly $d\sigma/dt$. The detector measures the kinematic properties of the outgoing muon, in particular its scattering angle and energy. Therefore, the differential cross section is usually expressed as a function of the scattering angle:

$$\frac{d\sigma}{d\Omega} = \frac{d\sigma}{dt}\left|\frac{dt}{d\Omega}\right|.$$

Since the momentum transfer is defined as

$$t = -Q^2,$$

(see Equation 3.12), the transformation factor can be written as

$$\frac{dt}{d\Omega} = -\frac{dQ^2}{d\Omega}.$$

For elastic muon-proton scattering in the laboratory frame, where the proton is initially at rest, the momentum transfer is given by

$$Q^2 = \frac{4E_\mu^2 \sin^2(\theta_\mu/2)}{1 + (2E_\mu/m_p)\sin^2(\theta_\mu/2)},$$

where θ_μ denotes the scattering angle of the outgoing muon. Using

$$d\Omega = 2\pi \sin\theta_\mu d\theta_\mu,$$

the transformation between the two differential cross sections becomes

$$\frac{dQ^2}{d\Omega} = \frac{1}{2\pi \sin\theta_\mu} \left| \frac{dQ^2}{d\theta_\mu} \right|.$$

The derivative of Q^2 with respect to the scattering angle can be calculated as

$$\frac{dQ^2}{d\theta_\mu} = \frac{2E_\mu^2 \sin(\theta_\mu)}{[1 + (2E_\mu/m_p)\sin^2(\theta_\mu/2)]^2}.$$

which leads to the final expression for the transformation factor:

$$\frac{dQ^2}{d\Omega} = \frac{E_\mu^2}{\pi[1 + (2E_\mu/m_p)\sin^2(\theta_\mu/2)]^2}.$$

Finally we have an expression for the differential cross section:

$$\frac{d\sigma}{d\Omega} = \frac{E_\mu^2}{16\pi^2\lambda(s, m_\mu^2, m_p^2)[1 + (2E_\mu/m_p)\sin^2(\theta_\mu/2)]^2} |\mathcal{M}|^2. \tag{3.13}$$

The remaining task is to determine the matrix element $\mathcal{M}$ describing the electromagnetic interaction.

3.4 Born Approximation

The starting point for calculating the scattering amplitude is the interaction part of the QED Lagrangian. At leading order in quantum electrodynamics (Born approximation), the interaction is described by the exchange of a single virtual photon.

$$\mathcal{L}_{\text{int}} = -e\overline{\psi}\gamma^\mu\psi A_\mu,$$

where ψ is the Dirac spinor field representing the charged fermion, A_μ is the electromagnetic four-potential, γ^μ are the Dirac gamma matrices, and e is the electric charge of the fermion. The interaction Lagrangian can therefore be written in terms of the electromagnetic current $j^\mu_{\text{em}} = \overline{\psi}\gamma^\mu\psi$ as

$$\mathcal{L}_{\text{int}} = -e j^\mu_{\text{em}} A_\mu.$$

This form makes the physical interpretation particularly transparent: the electromagnetic current couples to the electromagnetic field. In a scattering process, this coupling is represented diagrammatically by a vertex. The vertex therefore describes the interaction between a charged fermion and a photon. For a point-like fermion such as the muon, the electromagnetic vertex follows directly from the QED interaction Lagrangian. The corresponding Feynman rule is

$$-ie\gamma^{\mu},$$

and, written in terms of the incoming and outgoing muon spinors, the vertex factor is

$$\overline{u}(p_4)(-ie\gamma^{\mu})u(p_2),$$

where $u(p_2)$ and $\overline{u}(p_4)$ are the Dirac spinors for the incoming and outgoing muon, respectively.

For a composite particle like the proton, the electromagnetic vertex is modified by the internal structure of the particle. This modification is described by the electromagnetic form factors, which encode information about the spatial distribution of charge and magnetization within the proton. The vertex factor for the proton can be expressed as

$$\overline{u}(p_3)(-ie\Gamma^{\nu}(q^2))u(p_1),$$

where $u(p_1)$ and $\overline{u}(p_3)$ are the Dirac spinors for the incoming and outgoing proton, and $\Gamma^{\nu}(q^2)$ is the electromagnetic vertex function. It depends on the momentum transfer q^2 and can be expressed in terms of the Dirac and Pauli form factors $F_1(q^2)$ and $F_2(q^2)$:

$$\Gamma^{\nu}(q^2) = F_1(q^2)\gamma^{\nu} + \frac{i\sigma^{\nu\rho}q_{\rho}}{2m_p}F_2(q^2),$$

where $\sigma^{\nu\rho} = \frac{i}{2}[\gamma^{\nu}, \gamma^{\rho}]$ is the antisymmetric tensor constructed from the gamma matrices, and m_p is the mass of the proton. The Q^2 dependence of the form factors is shown in Figure 3.1. The Dirac form factor $F_1(q^2)$ describes the charge distribution, while the Pauli form factor $F_2(q^2)$ describes the anomalous magnetic moment of the proton. The Sachs form factors are related to the Dirac and Pauli form factors by

$$G_E(Q^2) = F_1(Q^2) - \frac{Q^2}{4m_p^2}F_2(Q^2)$$

and

$$G_M(Q^2) = F_1(Q^2) + F_2(Q^2).$$

The Q^2 dependence of the Sachs form factors is shown in Figure 3.2. The exchanged photon connects the muon and proton vertices. Since this photon is virtual, it is described by a photon propagator rather than by an external photon wave function. In Feynman gauge, the propagator is

$$\frac{-ig_{\mu\nu}}{q^2},$$

where $g_{\mu\nu}$ is the metric tensor. The muon vertex carries a free Lorentz index μ, while the proton vertex carries a free Lorentz index ν. Initially, these two indices are independent because they originate from two separate spinor chains, namely the lepton line and the hadron line. The two spinor chains cannot be multiplied directly; instead, they are connected

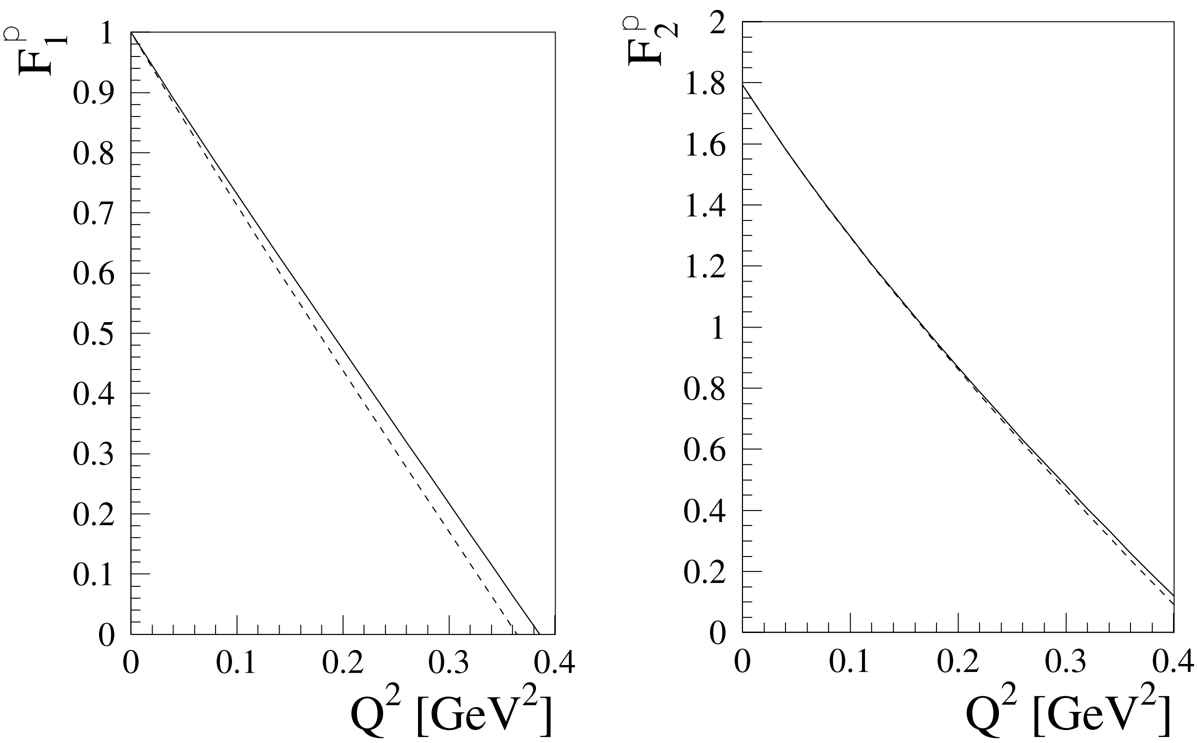

Figure 3.1: The Dirac (F_1) and Pauli (F_2) form factors of the proton in relativistic chiral perturbation theory at $\mathcal{O}(q4)$. Comparison of two different calculation approaches: EOMS (solid lines) and infrared regularization (dashed lines). Figures taken from Thomas Fuchs 2004.

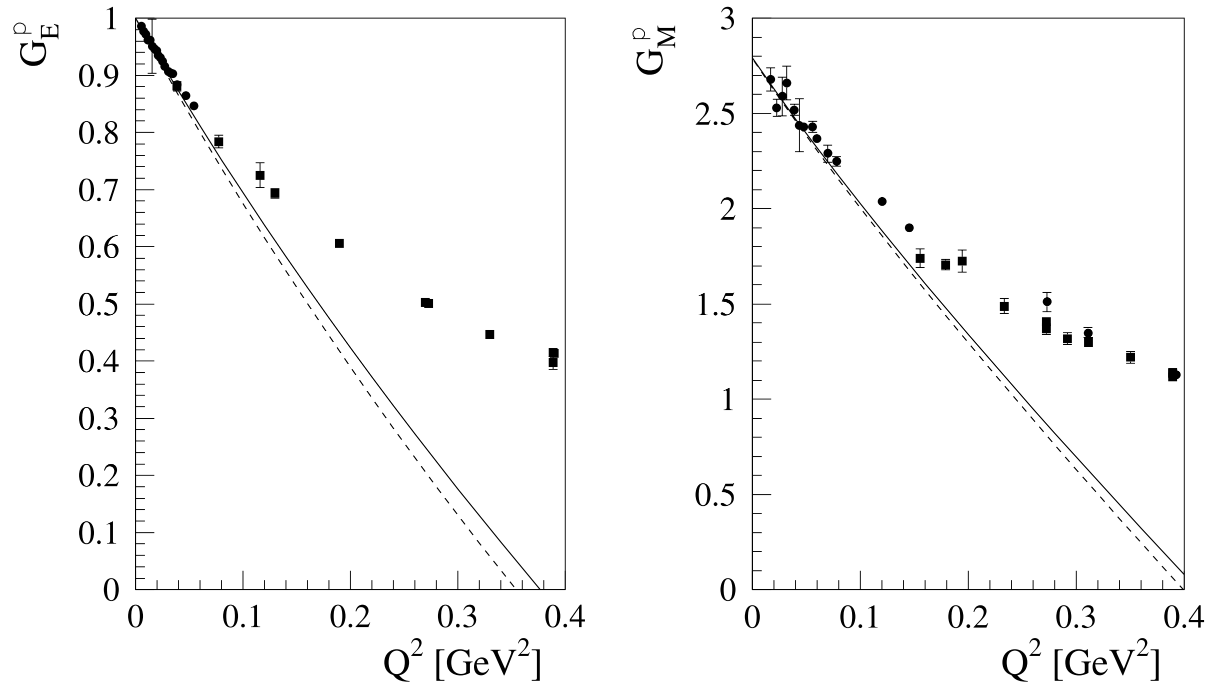

Figure 3.2: The Sachs form factors of the nucleon at $\mathcal{O}(q4)$. Comparison of two different calculation approaches: EOMS (solid lines) and infrared regularization (dashed lines). Figures taken from Thomas Fuchs 2004.

by the virtual-photon propagator. The propagator carries exactly two Lorentz indices, one associated with each side of the propagator, and the metric tensor $g_{\mu\nu}$ contracts these indices according to the Einstein summation convention.

Written with both Lorentz indices explicitly, the matrix element is

$$\mathcal{M} = [\overline{u}(p_4)(-ie\gamma^\mu)u(p_2)] \frac{-ig_{\mu\nu}}{q^2} \left[\overline{u}(p_3)(-ie\Gamma^\nu(q^2))u(p_1)\right].$$

The contraction with the metric tensor identifies the two Lorentz indices. In particular,

$$g_{\mu\nu}\Gamma^\nu(q^2) = \Gamma_\mu(q^2),$$

where the index is lowered according to the metric. Thus, the propagator effectively connects the Lorentz index of the leptonic current to that of the hadronic current. The matrix element can therefore be written as

$$\mathcal{M} = \frac{e^2}{q^2} \overline{u}(p_4)\gamma^\mu u(p_2)\, \overline{u}(p_3)\Gamma_\mu(q^2)u(p_1).$$

This expression is the starting point for calculating the cross section. However, the matrix element itself is not yet the observable quantity. The differential cross section is proportional to the squared modulus of the matrix element, summed over the unobserved final-state spins and averaged over the initial-state spins.

For unpolarized scattering, the spin-averaged squared matrix element is defined as

$$\overline{|\mathcal{M}|^2} = \frac{1}{4} \sum_{s_1,s_2,s_3,s_4} |\mathcal{M}|^2,$$

where the factor $1/4$ averages over the two possible initial spin states of the muon and the two possible initial spin states of the proton.

Writing the spin labels explicitly, the matrix element is

$$\mathcal{M} = \frac{e^2}{q^2} [\overline{u}(p_4,s_4)\gamma^\mu u(p_2,s_2)] \left[\overline{u}(p_3,s_3)\Gamma_\mu(q^2)u(p_1,s_1)\right].$$

Its complex conjugate is

$$\mathcal{M}^* = \frac{e^2}{q^2} [\overline{u}(p_2,s_2)\gamma^\nu u(p_4,s_4)] \left[\overline{u}(p_1,s_1)\overline{\Gamma}_\nu(q^2)u(p_3,s_3)\right],$$

where

$$\overline{\Gamma}_\nu = \gamma^0\Gamma_\nu^\dagger\gamma^0.$$

Therefore,

$$|\mathcal{M}|^2 = \frac{e^4}{(q^2)^2} [\overline{u}(p_4)\gamma^\mu u(p_2)] [\overline{u}(p_2)\gamma^\nu u(p_4)] \left[\overline{u}(p_3)\Gamma_\mu u(p_1)\right] \left[\overline{u}(p_1)\overline{\Gamma}_\nu u(p_3)\right].$$

To evaluate the spin sums, the completeness relation for Dirac spinors is used:

$$\sum_s u(p,s)\overline{u}(p,s) = \not{p} + m,$$

where

$$\not{p} = \gamma^\mu p_\mu.$$

This relation allows the products of spinors to be converted into traces over products of gamma matrices. The spin-averaged squared matrix element can consequently be written as

$$\overline{|\mathcal{M}|^2} = \frac{e^4}{4(q^2)^2} L^{\mu\nu} H_{\mu\nu},$$

where the leptonic tensor is

$$L^{\mu\nu} = \mathrm{Tr}\left[(\not{p}_4 + m_\mu)\gamma^\mu(\not{p}_2 + m_\mu)\gamma^\nu\right],$$

and the hadronic tensor is

$$H_{\mu\nu} = \mathrm{Tr}\left[(\not{p}_3 + m_p)\Gamma_\mu(q^2)(\not{p}_1 + m_p)\overline{\Gamma}_\nu(q^2)\right].$$

The factorization into a leptonic and a hadronic tensor separates the two physical contributions. The leptonic tensor describes the coupling of the virtual photon to the muon, while the hadronic tensor contains the proton electromagnetic structure through the form factors.

For the point-like muon, the leptonic tensor can be evaluated using the standard gamma-matrix trace identities. In the limit where the muon mass can be neglected, this gives

$$L^{\mu\nu} = 4\left(p_4^\mu p_2^\nu + p_4^\nu p_2^\mu - g^{\mu\nu} p_4 \cdot p_2\right).$$

The hadronic tensor contains the proton vertex $\Gamma^\mu(q^2)$ and therefore the form factors F_1 and F_2. Evaluating the hadronic trace and contracting it with the leptonic tensor gives the spin-averaged squared matrix element in terms of the kinematic variables and the proton form factors.

The resulting quantity is then inserted into the general expression for the differential cross section Equation 3.13. Evaluating the leptonic and hadronic tensors and performing the contraction $L^{\mu\nu}H_{\mu\nu}$ then leads to the Rosenbluth form of the elastic Born cross section.

$$\frac{d\sigma_{\mathrm{Born}}}{d\Omega} = \frac{d\sigma_{\mathrm{Mott}}}{d\Omega} \frac{\tau}{\epsilon(1+\tau)} \left(G_M^2(Q^2) + \frac{\epsilon}{\tau} G_E^2(Q^2)\right), \tag{3.14}$$

where $\frac{d\sigma_{\mathrm{Mott}}}{d\Omega}$ isthe Mott cross section which describes the elastic scattering of a charged lepton from a structureless charged target. In the ultrarelativistic limit, it is given by

$$\frac{d\sigma_{\mathrm{Mott}}}{d\Omega} = \frac{\alpha^2 \cos^2(\theta/2)}{4E^2 \sin^4(\theta/2)}. \tag{3.15}$$

Moreover,

$$\tau = \frac{Q^2}{4m_p^2} \tag{3.16}$$

is a dimensionless parameter and

$$\epsilon = \left(1 + 2(1+\tau)\tan^2\frac{\theta}{2}\right)^{-1} \tag{3.17}$$

is the so-called virtual photon polarization parameter.

The internal structure of the proton is described by the electric and magnetic Sachs form factors. In the low-Q^2 region relevant for proton radius measurements, these form factors can be approximated by the empirical dipole parametrization

$$G_D(Q^2) = \left(1 + \frac{Q^2}{\Lambda^2}\right)^{-2}, \tag{3.18}$$

where the dipole mass parameter is

$$\Lambda^2 = 0.71\,\mathrm{GeV}^2. \tag{3.19}$$

The electric and magnetic form factors are then expressed as

$$G_E(Q^2) = G_D(Q^2) = \left(1 + \frac{Q^2}{0.71\,\mathrm{GeV}^2}\right)^{-2},^{1} \tag{3.20}$$

and

$$G_M(Q^2) = \mu_p G_D(Q^2) = \mu_p\left(1 + \frac{Q^2}{0.71\,\mathrm{GeV}^2}\right)^{-2}, \tag{3.21}$$

where $\mu_p = 2.792847$ is the dimensionless proton magnetic moment in units of the nuclear magneton μ_N. It defines the normalization of the magnetic form factor, such that $G_M(0) = \mu_p$.

3.5 Radiative Corrections

Relevant for this thesis are radiative corrections to next to leading order (NLO). These include processes with one additional photon. The corresponding Feynman diagrams are shown in Figure 3.3. The radiative corrections can be divided into two categories: virtual corrections and real photon emission.

3.5.1 Virtual Photons

The virtual corrections arise from loop diagrams, where a virtual photon is emitted and reabsorbed by the charged particles. These diagrams modify the interaction vertex and the propagators of the particles involved.

[1] For a point-like proton, the form factor would be $G_E(Q^2) = 1$ for all Q^2.

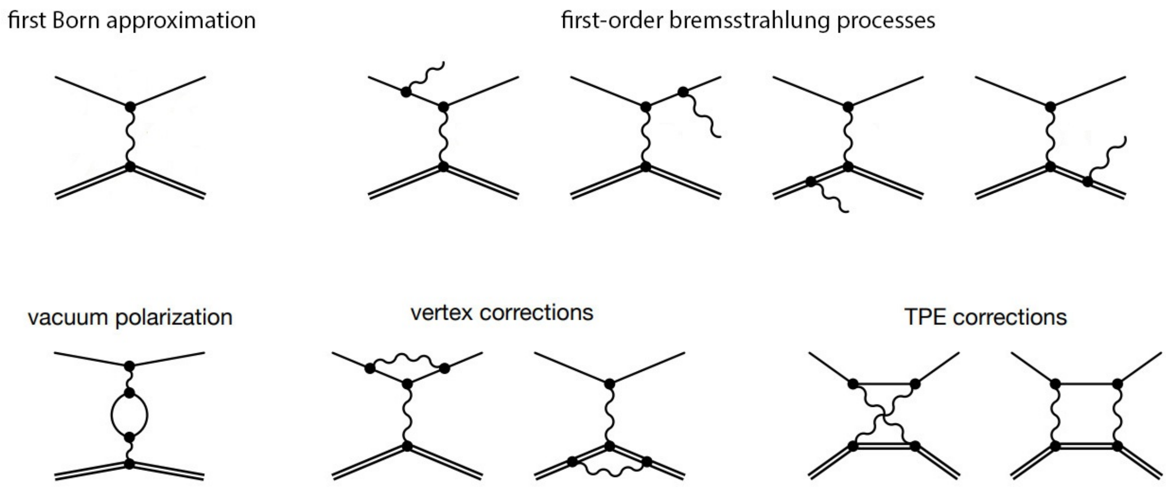

Figure 3.3: LO and NLO processes. Figure taken from Mehran Depour.

3.5.2 Real Photon Emission

The real photon emission corresponds to processes where an additional photon is emitted in the final state.

4 Simulation with McMule

4.1 Overview of McMule

4.2 Simulation Setup and Implementation of the ECal2 Detector

5 Event Generation with McMule

6 Results

6.1 Comparison of Q^2 distribution with Analytical solution

6.2 Impact of radiative corrections

6.3 Influence of Analysis Cuts

6.3.1 The Scattered Muon Angle Range

6.3.2 The Photon Energy Threshold

7 Discussion

- Are radiative corrections significant for AMBER?
    - Which phase-space regions are most affected?
    - Are detector cuts sufficient?
    - What precision is required?

8 Conclusion and Outlook

ECal2 geht nicht... sad.

List of Acronyms

PRM proton radius measurement

QED quantum electrodynamics

TPC Time Projection Chamber

Bibliography

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